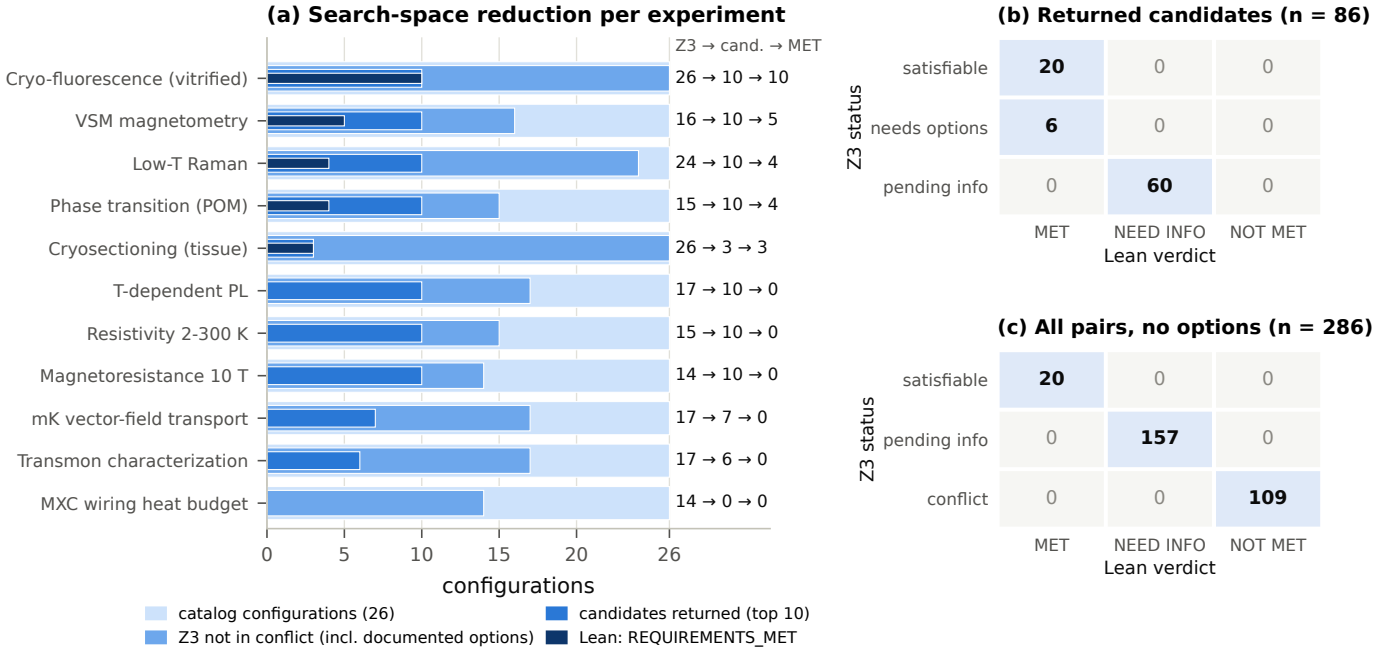
(a) Search-space reduction per experiment
Z3 → cand. → MET
Cryo-fluorescence (vitrified)
26 → 10 → 10
VSM magnetometry
16 → 10 → 5
Low-T Raman
24 → 10 → 4
Phase transition (POM)
15 → 10 → 4
Cryosectioning (tissue)
26 → 3 → 3
T-dependent PL
17 → 10 → 0
Resistivity 2-300 K
15 → 10 → 0
Magnetoresistance 10 T
14 → 10 → 0
mK vector-field transport
17 → 7 → 0
Transmon characterization
17 → 6 → 0
MXC wiring heat budget
14 → 0 → 0
0
5
10
15
20
26
configurations
catalog configurations (26)
Z3 not in conflict (incl. documented options)
candidates returned (top 10)
Lean: REQUIREMENTS_MET
(b) Returned candidates (n = 86)
Z3 status
satisfiable
needs options
pending info
20
0
0
6
0
0
0
60
0
MET
NEED INFO
NOT MET
Lean verdict
(c) All pairs, no options (n = 286)
Z3 status
satisfiable
pending info
conflict
20
0
0
0
157
0
0
0
109
MET
NEED INFO
NOT MET
Lean verdict